

A.L.I.V.E. Active Leash for Intelligent Vigilance & Ease

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Motivation

Jogging with your dog should feel smooth and safe, but passive leashes cannot absorb both sudden pulls or control extra length simultaneously.

+ 20'000 injuries/year caused by dog-leashes in USA^[1,2]

Causes : - sudden pull from the dog
- tripping or tangling in the leash

This motivated the development of **A.L.I.V.E. !**

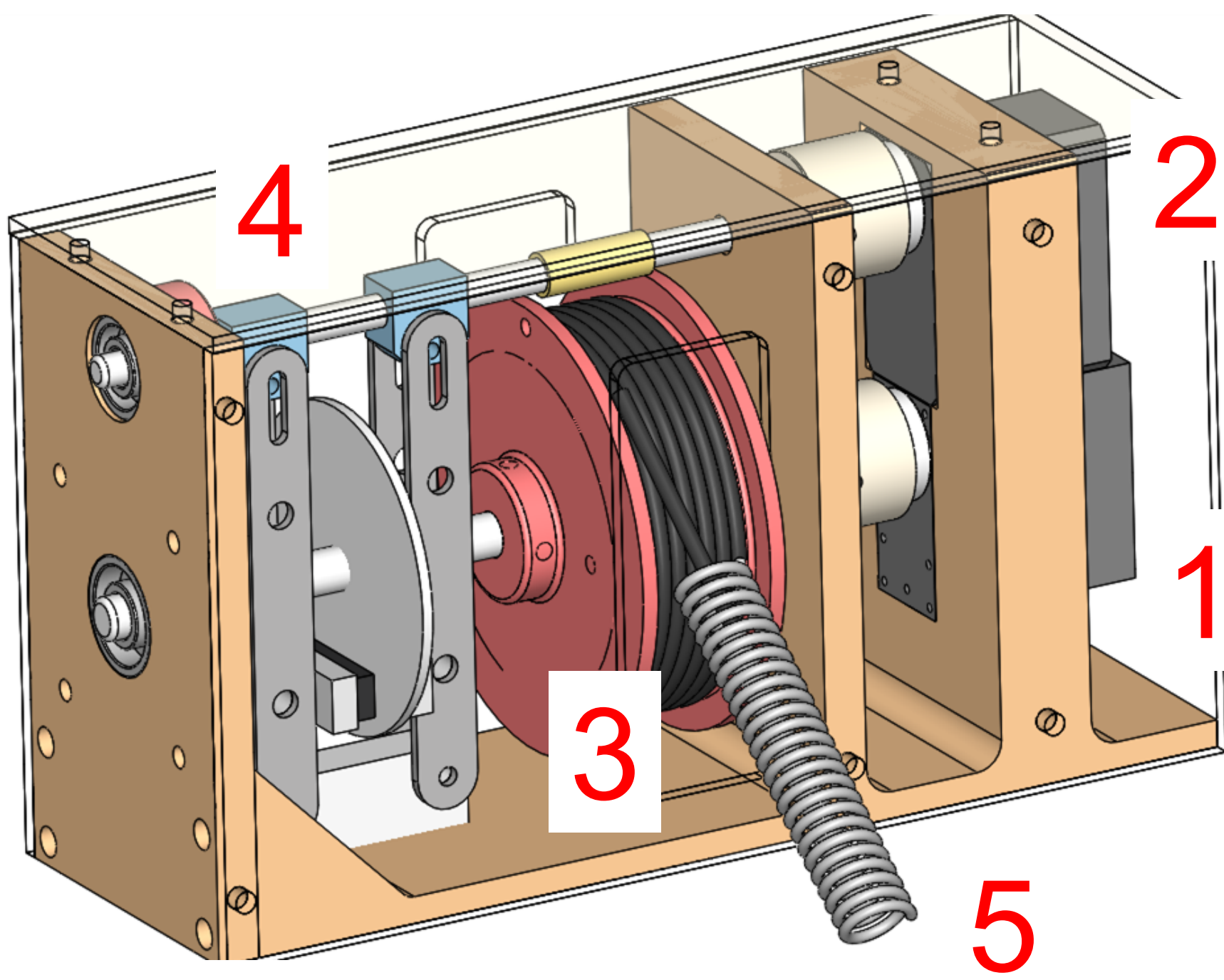
Functionality

The user sets a desired leash length!

- Dog's position past desired length: A.L.I.V.E. applies smooth braking to damp the pull, then reels the dog back to the set length
- Dog's position within desired length: keeps the leash taut and off the ground by winding backward

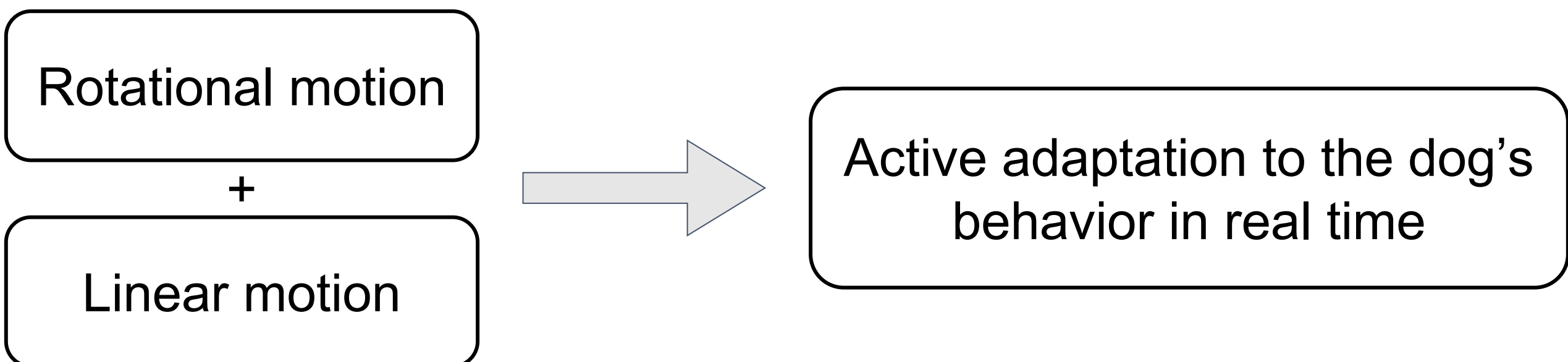
Design

1. Primary motor
2. Secondary motor
3. Rotating spool
4. Braking system
5. Leash's spring



Mechanical

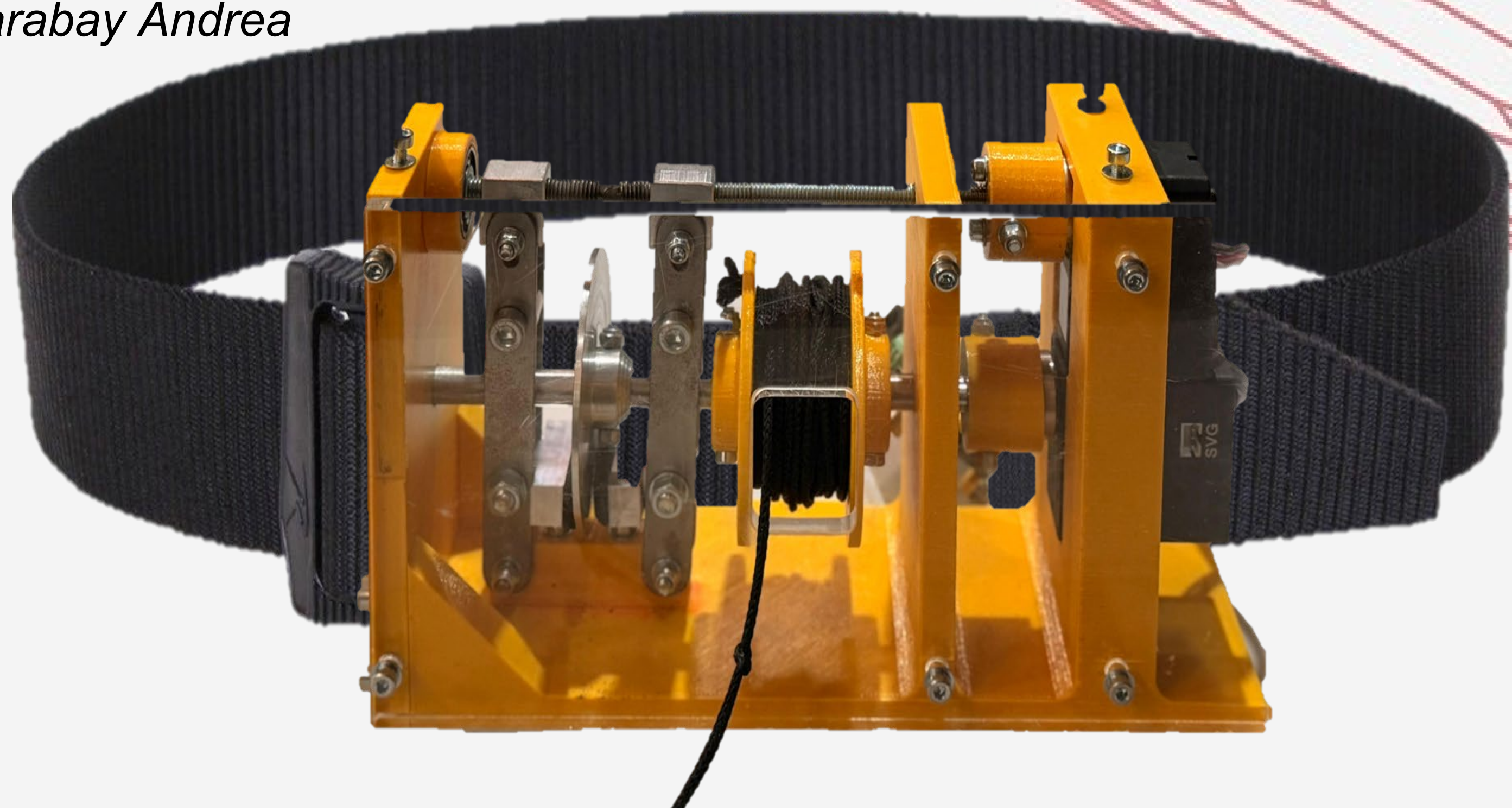
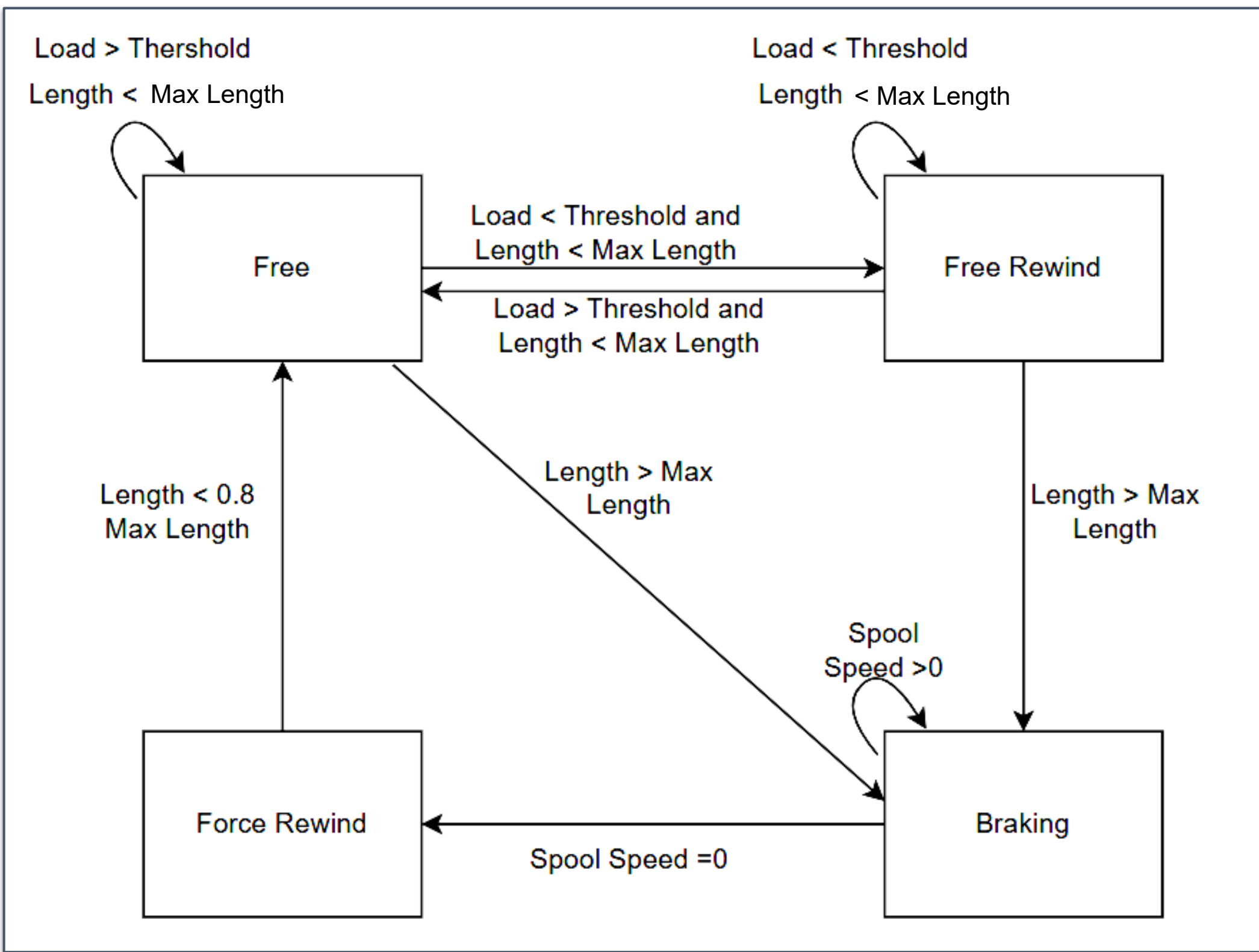
Overall principle



- 1. Rotating spool**
 - Driven by the main motor
 - Winds the leash in or out
 - Control of the length
 - Maintains tension so the leash does not touch the ground
- 2. Slowing / braking system**
 - Driven by the second motor
 - Linear mechanism
 - engages a brake
 - Slows down the main axis
 - Smooth handling of sudden pulls

Resulting behavior : With the spring, the two actuators manage both leash length and pull force, keeping tension smooth during accelerations and decelerations.

Control Logic



State-of-the-art

Most leashes are passive.



Retractable Leash:
No shock absorption, sudden pulls hit the person directly

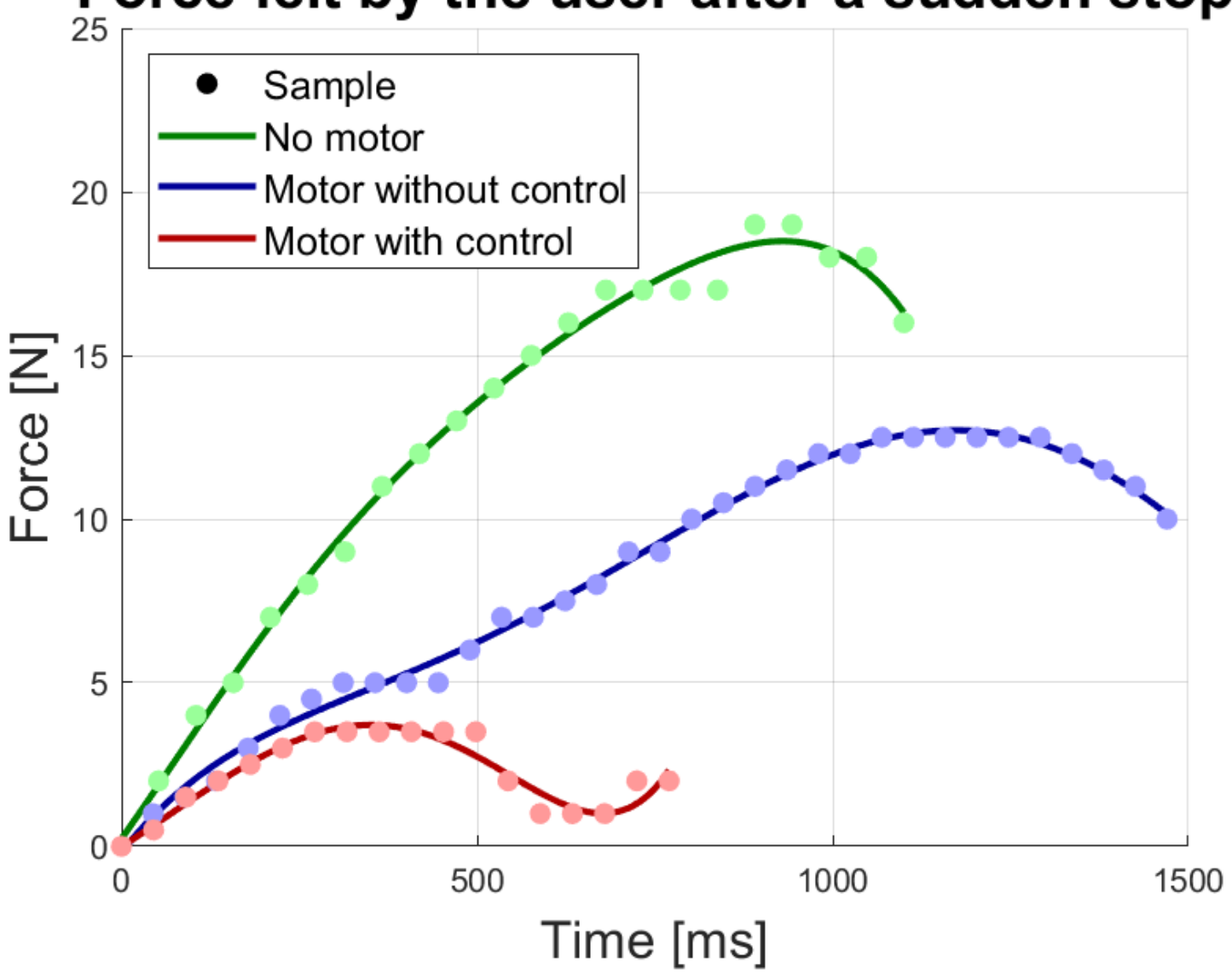


Bungee / Waist Leash:
When leash is extended, it drags & can't reel the dog back.

A.L.I.V.E. tackles both problems by
1. softening sudden pulls 2. automatically retracting the extra rope

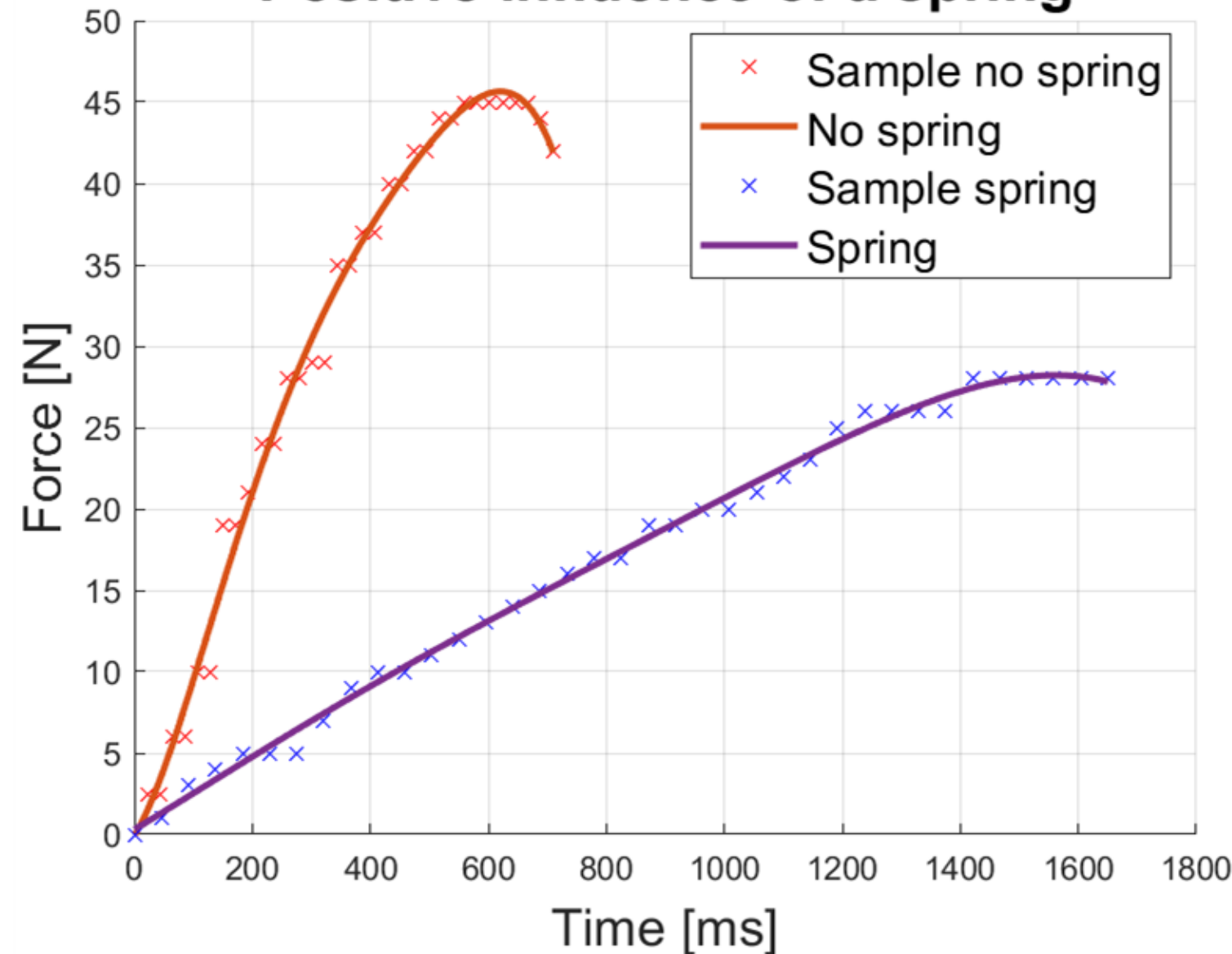
Performance

Force felt by the user after a sudden stop



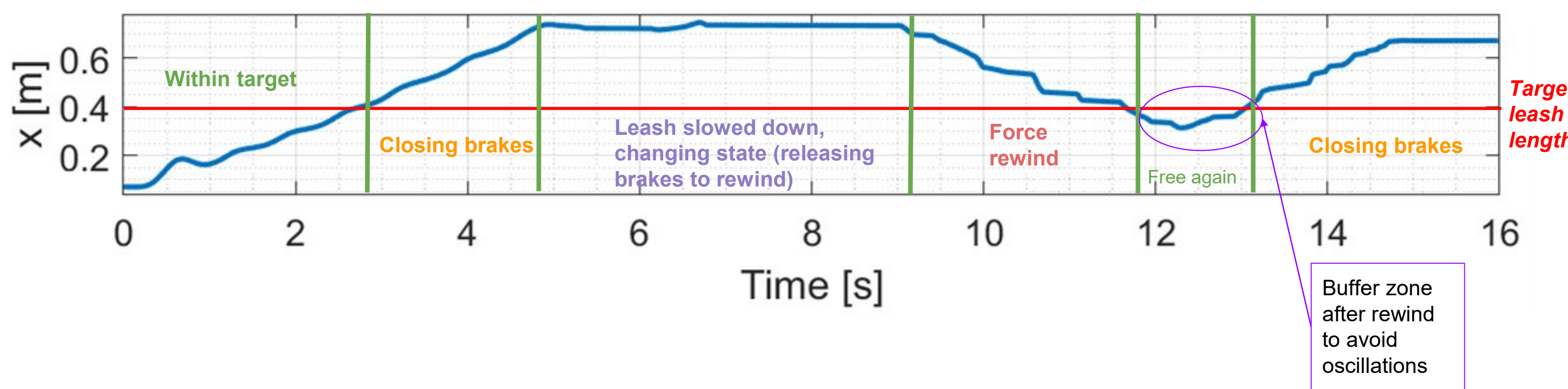
Setup: Mass of 1kg with an acceleration of ~3 m/s²
A spring is put between the mass and the end of the rope

Positive influence of a spring



Analysis: The force experienced by a 1kg mass in free fall after sudden stop is lower by using a spring

Position tracking of a Mass Passing the Desired Leash Length

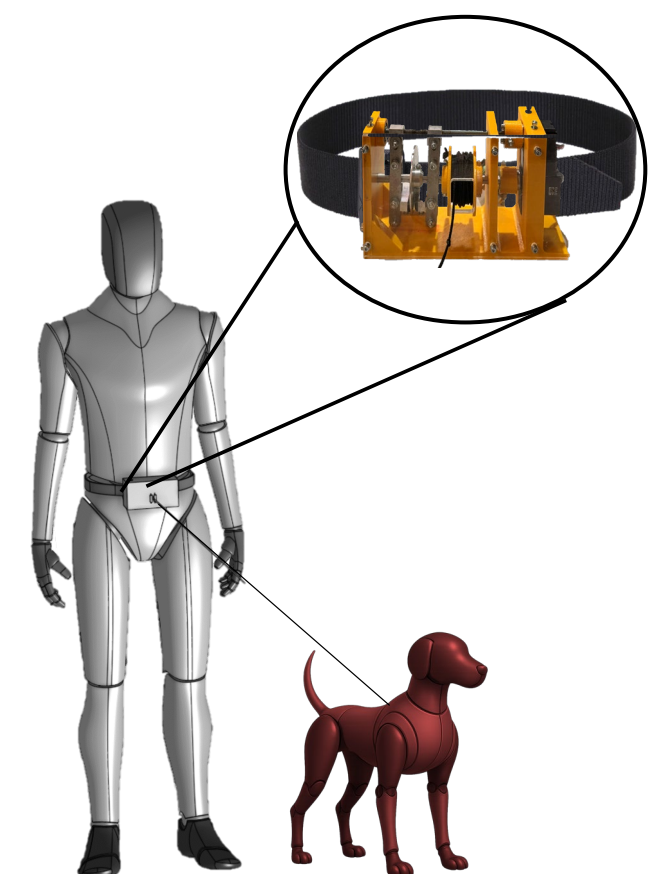


Future Development

A.L.I.V.E. shows that an active two-motor leash can control both length and pull force to improve safety and comfort.

Next steps:

- Make the device more compact and lighter.
- Replace the spool servo (initially chosen for budget reasons) with a more backdrivable clutch-based mechanism.
- Add a railing system to redirect leash tension when the dog changes position.



References :

1. AVMA Publication, Leash-related injuries associated with dog walking: an understudied risk for dog owners?, Rodrigo rosa and Richard E. Buckley, Apr 2024
2. National library of medicine, Dog leash-related injuries treated at emergency departments, Mathias B. Forrester, 05.2020